

Processes Cheat Sheet (ps, top, kill)

See what's running, what's using the CPU and memory, and stop programs that misbehave.

Using it

`ps aux` · `ps -ef` · `top` ← all processes · all, with parent PID (PPID) · live view (q quits)

Reading ps aux

USER	PID	%CPU	%MEM	VSZ	RSS	TTY	STAT	START	TIME	COMMAND
root	1	0.0	0.1	167748	12960	?	Ss	Oct02	0:05	/sbin/init ← s = session leader
mysql	911	2.1	12.4	2482036	498212	?	Ssl	Oct02	31:07	/usr/sbin/mysqld
mark	4410	0.0	0.1	10072	5308	pts/0	Ss	09:12	0:00	-bash
mark	4532	99.6	0.0	5572	1004	pts/0	R+	09:30	2:14	yes ← CPU hog · + = foreground

Column	Means	STAT	Means
%CPU	Share of one CPU (can pass 100 if threaded)	R	Running or ready to run
%MEM	Share of total RAM	S	Sleeping, waiting for something (normal)
RSS	Real RAM in use, in KB (VSZ : reserved; ignore)	D	Stuck waiting on disk/network; can't be killed
TTY	Its terminal (? = none: a service)	Z	Zombie: dead, parent hasn't cleaned up
TIME	Total CPU time used so far	T	Stopped (paused with Ctrl+Z or STOP)

Finding and stopping processes

Command	Does	Signal	Use it to
pgrep -a nginx	PIDs and command lines matching nginx	15 TERM	Ask it to quit cleanly (default)
kill 4532	Sends TERM to PID 4532	2 INT	Interrupt, same as Ctrl+C
kill -9 4532	Sends KILL: instant, no cleanup	1 HUP	Many daemons reload their config
pkill nginx	Sends TERM to every match by name	9 KILL	Force it dead. Last resort
pkill -f app.py	Matches the full command line	19 / 18	STOP pauses it / CONT resumes it

Reading top

```
top - 09:31:02 up 1 day, 2:16, 2 users, load average: 1.12, 0.84, 0.61
%Cpu(s): 24.3 us, 1.2 sy, 0.0 ni, 73.9 id, 0.4 wa, 0.0 hi, 0.2 si, 0.0 st
MiB Mem : 7841.6 total, 912.4 free, 2826.9 used, 4102.3 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used. 4720.8 avail Mem
```

You see	Means	Key	Does
load average	How busy, over 1, 5, 15 min. Worry if above nproc	P / M	Sort by CPU / by memory
us / sy / id	CPU used by programs / by the kernel / idle	1	Show each CPU separately
wa	Waiting on disk. High = storage is the bottleneck	c	Show full command lines
avail Mem	RAM really free for new programs	k / q	Kill (asks for PID) / quit

Priority, background jobs, open files

Command	Does	Command	Does
nice -n 10 cmd	Lower priority (-20 high ... 19 low)	cmd &	Runs cmd in the background
renice -n 5 -p 4532	New priority for a running PID	Ctrl+Z, bg	Pause job, resume it in background
lsof -p 4532	Files and sockets PID 4532 has open	jobs, fg %1	List jobs / bring job 1 to the front
sudo lsof -i :80	Which process is using port 80	nohup cmd &	Keeps running after you log out

Handy recipes

Command	Does
ps aux --sort=-%mem head	Top memory users (-%cpu for CPU)
ps -o ppid= -p 4532	Parent of a process (kill the parent to clear zombies)

Remember: try plain **kill** first and wait a few seconds; **kill -9** gives the program no chance to save work or clean up. Nothing kills a process stuck in state **D**.